STRICTNESS IS THE COINCIDENCE OF THE QUOTIENT FILTRATION WITH THE INDUCED FILTRATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; both filtrations are identified through named preceding results and the closing sentence is a conclusion rather than a placeholder.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- The canonical morphism from the coimage to the image. Same chapter, Lemma lemma-coim-im-map. Supplies the morphism whose being an isomorphism of filtered objects is the second condition of the lemma.
- Kernels and cokernels of filtered objects. Same chapter, Lemma lemma-filtered. Supplies the description of the filtration on the coimage as a quotient filtration and on the image as an induced filtration, which is what the proof compares.

Source reference. Original lemma 0107: sources/stacks-homology/pinned-original-source.tex, source lines 515–515. Navigation pointer invoked at proof line 4976; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0122: sources/stacks-homology/pinned-original-source.tex, source lines 4928–4928. Navigation pointer invoked at proof line 4924; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:homology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $f : A \to B$ be a morphism of filtered objects of $\mathcal{A}$. The following are equivalent*

(1) *f is strict,*
(2) *the morphism $\mathrm{Coim}(f) \to \mathrm{Im}(f)$ of Lemma 0107 is an isomorphism.*

Proof. Note that $\mathrm{Coim}(f) \to \mathrm{Im}(f)$ is an isomorphism of objects of $\mathcal{A}$, and that part (2) signifies that it is an isomorphism of filtered objects. By the description of kernels and cokernels in the proof of Lemma 0122 we see that the filtration on $\mathrm{Coim}(f)$ is the quotient filtration coming from $A \to \mathrm{Coim}(f)$. Similarly, the filtration on $\mathrm{Im}(f)$ is the induced filtration coming from the injection $\mathrm{Im}(f) \to B$. The definition of strict is exactly that the quotient filtration is the induced filtration. □